

Vapor–Liquid Equilibria in Binary Systems of Phenol or Cresols + Water, + Toluene, and + Octane and Liquid–Liquid Equilibria in Binary Systems of Cresols + Water

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Isothermal vapor–liquid equilibrium data were measured at 333.15 and 363.15 K for the systems toluene + phenol, toluene + 2-cresol, toluene + 3-cresol, toluene + 4-cresol, octane + 2-cresol, octane + 3-cresol, water + phenol, and water + 3-cresol. Additionally, heteroazeotropic data of water + 2-cresol and water + 4-cresol were determined. The solubility of water in 2-cresol, 3-cresol, and 4-cresol was measured by photometric turbidity titration. Excess molar volumes for the systems toluene + phenol or cresols and octane + 2-cresol or + 3-cresol are presented together with Redlich–Kister polynomial fits. The VLE data and, if present, LLE data have been regressed according to the models NRTL, UNIQUAC, and Elliott–Suresh–Donohue equation of state together with data from the literature.

Introduction

In a recent published report¹ we presented parameters for the models NRTL,² UNIQUAC,³ and the Elliott–Suresh–Donohue equation of state^{4,5} (ESD EOS). These are based on various VLE and LLE data measured in our group that are not published yet. Those data are presented in this work together with a revised parameter table. Isothermal VLE data were measured at 333.15 and 363.15 K for the homogeneous systems toluene + phenol, toluene + 2-cresol, toluene + 3-cresol, toluene + 4-cresol, octane + 2-cresol, and octane + 3-cresol and for the systems water + phenol and water + 3-cresol only in the homogeneous concentration range. Additionally, heteroazeotropic data for water + 2-cresol and water + 4-cresol were determined at 333.15–363.15 K in steps of 10 K. The VLE samples were analyzed by measurement of density, or if water was present by Karl Fischer titration. The densities for the systems toluene + phenol, + 2-cresol, + 3-cresol, and + 4-cresol, and octane + 2-cresol and + 3-cresol were determined and excess molar volumes were calculated. The solubility of water in 2-cresol, 3-cresol, and 4-cresol was measured. These data have been regressed together with data from the literature according to the models NRTL, UNIQUAC, and ESD EOS.

Experimental Section

Materials. Phenol (VEB Berlin Chemie, Berlin Germany, purity > 98%), 2-cresol (Aldrich Chemical, Steinheim, Germany, purity > 98%), and 3- and 4-cresol (Merck-Schuchardt, Hohenbrunn, Germany, purity > 99%) were distilled twice in a Vigreux column at reduced pressure under a N₂ atmosphere. Octane (purity > 99%) and toluene (purity p.a.) were obtained from Arcros Organics, Belgium, and were also distilled in a Vigreux column at reduced pressure. The refractive indices and the densities of pure substances are in agreement with literature values (listed in Table 1). 2-Cresol, 3-cresol, 4-cresol, octane, and toluene were dried over sodium sulfate. Purities determined by gas liquid chromatography were as follows: phenol > 99.7%,

2-cresol > 99.1%, 3-cresol > 99.9%, 4-cresol > 99.8%, toluene > 99.8%, and octane > 99.9%. Deionized and distilled water was used.

Apparatus and Procedure. VLE were determined in a Röck and Sieg type circulation still. The apparatus and experimental setup is described in detail by Schmelzer et al.⁸ Since in some systems the condensed vapor phase was heterogeneous, a different Röck and Sieg type circulation still was used (cp. Figure 1). The principle of the circulation still is briefly described here. The boiling flask (1) contains about 250 cm³ of liquid mixture. Vapor and liquid pass through a Cottrell tube (2). Vapor arises and is condensed in the cooling part (3), passes the three-way stopcock (4) and a mixing vessel (5), and flows back to the boiling flask (1). Liquid flows by a liquid sample vessel (6) and from there to a mixing vessel (5) and finally back into the boiling flask (1). If equilibrium is reached, the stopcock plug (4) made of PTFE is opened so that condensed vapor is sampled into test tubes. Splitting of the liquid phases in the boiling flask in cases of heterogeneous liquid composition is prevented by intense stirring.

The liquid and vapor samples were analyzed by Karl Fischer titration with a 787 KF Titrimo (Deutsche Metrohm, Filderstadt) if one component was water or otherwise by measurement of density. For this task densities over the entire concentration range for the systems toluene + phenol, + 2-cresol, + 3-cresol, and + 4-cresol and octane + 2-cresol or + 3-cresol were determined. The densities of gravimetrically prepared samples were measured with a vibrating tube density meter (DMA 58,

Table 1. Experimental and Literature Values of Refractive Indices and Densities of Pure Components

substance	T (K)	n_D		ρ (g·cm ³)	
		exptl	lit ⁶	exptl	lit ⁶
phenol	313.15	1.5409	1.5408	1.05830	1.05877
2-cresol	308.15	1.5386	1.5386	1.03273	1.0327
3-cresol	293.15	1.5401	1.5401	1.03385	1.0339
4-cresol	313.15	1.5311 ^a	1.53115 ^{a,7}	1.01880	1.0185
octane	298.15	1.3946	1.3944	0.69849	0.6986
toluene	293.15	1.4960	1.4961	0.86678	0.8668

^a At 314.15 K.

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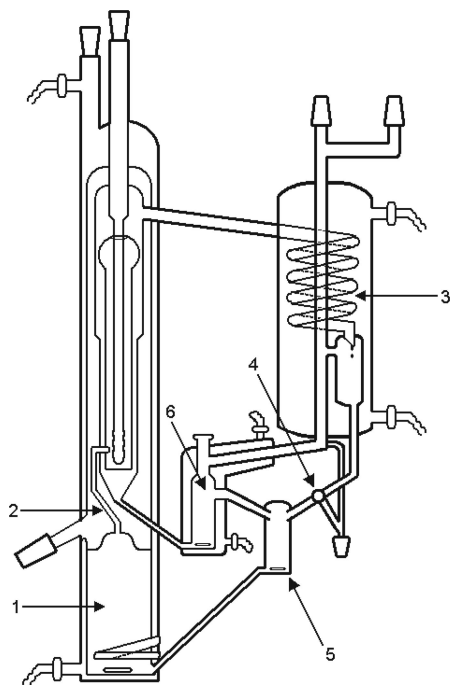


Figure 1. Modified Röck and Sieg type circulation still: 1, boiling flask; 2, Cottrell tube; 3, cooling part; 4, three-way stopcock with PTFE plug; 5, mixing vessel; 6, liquid sample vessel.

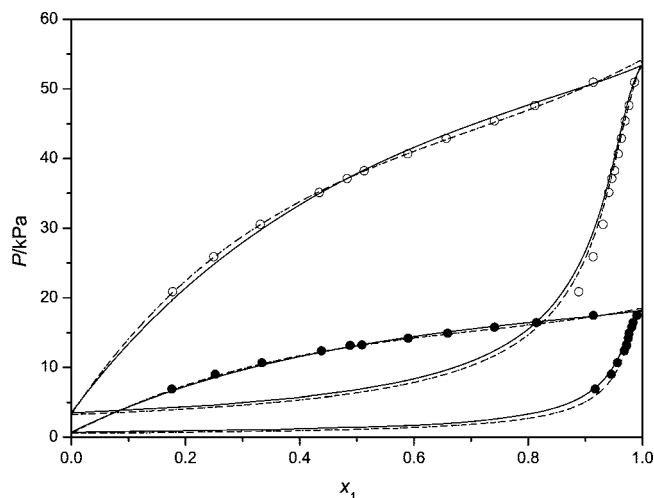


Figure 2. VLE for toluene (1) + phenol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

Anton Paar, Austria). The accuracy of the density meter was $\Delta\rho/\text{g}\cdot\text{cm}^{-3} = 0.00002$ with $\Delta T/\text{K} = 0.01$ and $\Delta x = 0.0002$. The absolute errors of VLE measurement were less than the following: for pressure $\Delta P/\text{kPa} = 0.06$, for temperature $\Delta T/\text{K} = 0.05$, and for composition (mole fraction) $\Delta x = \Delta y = 0.005$.

The solubility of water in cresols was determined by photometric turbidity titration method. The experimental buildup is described in Schmelzer et al.¹ and Klauck et al.⁹ The absolute errors of liquid-phase determination were less than $\Delta x = 0.005$ and for temperature $\Delta T/\text{K} = 0.1$.

Results

The results of VLE/VLLE measurements are presented in Figures 213 and are given in the Supporting Information. All VLE show positive deviations from Raoult's law whereby repulsive forces between phenols and the second component

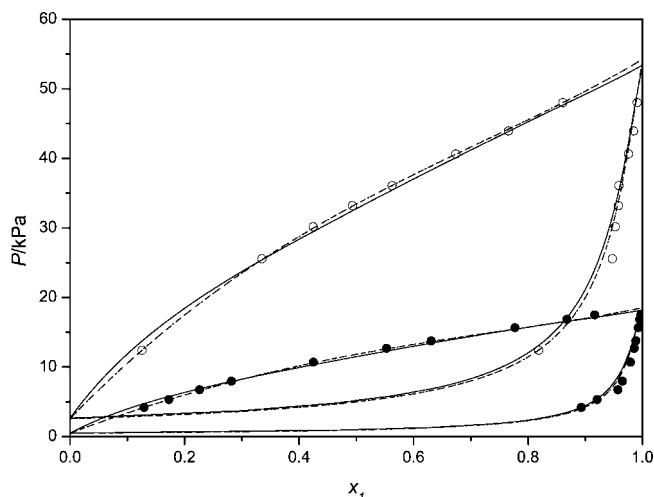


Figure 3. VLE for toluene (1) + 2-cresol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

increase in the order toluene, octane, and water. All systems have a great difference in pure component vapor pressure; therefore, the vapor composition mainly consists of low-boiling substance. The water-containing systems are only partially miscible and show an upper critical solution temperature.

The measured P – x – y data were tested for consistency with the method of Christiansen and Fredenslund.¹⁰ Vapor pressure data for all substances were calculated from the DIPPR correlation.¹¹ The coefficients for a Legendre polynomial were determined by minimizing deviations of total pressures.

$$\frac{G^E}{RT} = x_1 x_2 \sum_k A_k L_k(x_1)$$

$$L_k(x_1) = [(2k-1)(x_1 - x_2)L_{k-1}(x_1) - (k-1)L_{k-2}(x_1)]/k$$

$$L_0(x_1) = 1 \quad L_1(x_1) = x_1 - x_2 \quad (1)$$

The consistency criterion is defined by

$$|y_i^{\text{calcd}} - y_i^{\text{exptl}}| < \Delta x_i + \Delta y_i \quad (2)$$

Upon recommendation by Danner and Gess,¹² the deviation of experimental and calculated vapor composition should be less than 0.01. The deviations for a Legendre polynomial with four constants are shown in Table 2 and are less than 0.7% for total pressure and less than 0.009 for vapor composition for all systems. All measured VLE data were also tested by the method proposed by Van Ness.¹³ Thereby, the ratios of experimental and calculated activity coefficients (for example, from Legendre polynomial) are compared for each data point. Since the vapor composition in some of the investigated systems is almost pure low-boiling component, the experimental activity coefficient is very sensitive to deviations of vapor composition. This leads to poor classification of some systems in the categories introduced by Van Ness. For this reason the consistence test of Christiansen and Fredenslund was assumed to be more convincing proof of consistency for the investigated systems.

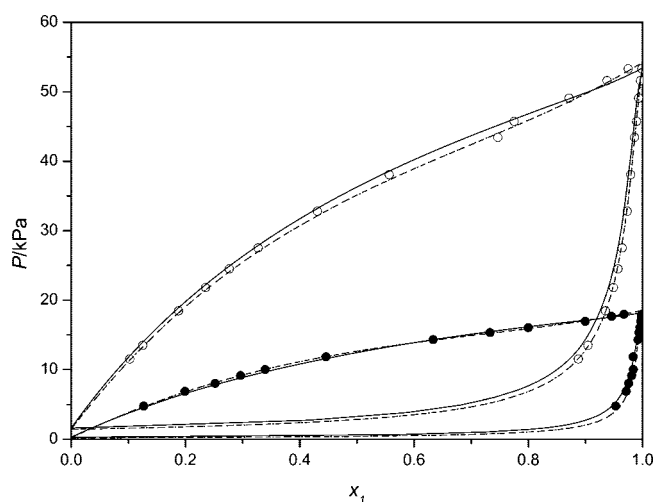
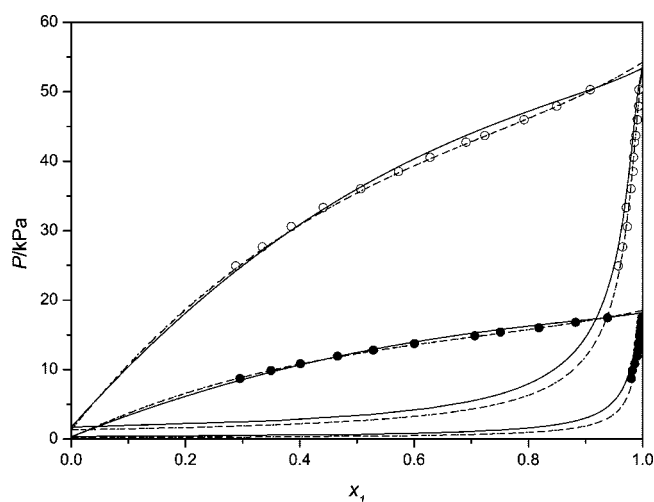
Densities were measured at 303.15 K in the systems toluene + 2-cresol, + 3-cresol, and + 4-cresol and in octane + 2-cresol and + 3-cresol and at 313.15 K in the system toluene + phenol. The results are available in the Supporting Information.

The results of solubility measurement are shown in Figure 14 and given in the Supporting Information. The systems water + phenol or cresols show increasing solubility with increasing temperature. The hydrophobicity increases notably from phenol to the cresols and for the cresols in the order 4-cresol, 3-cresol,

Table 2. Coefficients of Legendre Polynomial (eq 1) and Deviations of Experimental and Calculated VLE Data

system	<i>T</i> (K)	<i>A</i> ₁	<i>A</i> ₂	<i>A</i> ₃	<i>A</i> ₄	Δ <i>P</i> ^a (%)	Δ <i>y</i> ^a
toluene + phenol	333.15	1.0485	0.3867	0.0152	0.0783	0.29	0.0015
	363.15	0.9936	0.1711	0.0286	0.0234	0.11	0.0083
toluene + 2-cresol	333.15	0.7133	0.3508	0.0396	0.0867	0.51	0.0058
	363.15	0.5904	0.1078	−0.0078	0.0013	0.16	0.0066
toluene + 3-cresol	333.15	1.0202	0.2570	0.0757	0.0097	0.33	0.0012
	363.15	0.8741	0.1702	0.0169	0.0019	0.49	0.0023
toluene + 4-cresol	333.15	0.8835	0.3016	0.0272	0.0117	0.16	0.0012
	363.15	0.7794	0.3027	−0.0266	0.0304	0.15	0.0019
octane + 2-cresol	333.15	1.9349	0.1515	0.1486	0.0608	0.34	0.0028
	363.15	1.8241	−0.0719	0.1341	−0.0304	0.18	0.0059
octane + 3-cresol	333.15	1.6722	0.3753	−0.0320	−0.0025	0.49	0.0021
	363.15	2.0074	0.0353	0.1352	−0.0394	0.56	0.047
water + 3-cresol	333.15	1.7342	1.6300	0.2856	0.3632	0.65	0.0002
	363.15	1.6533	1.5910	0.3790	0.3343	0.46	0.0045

^a Δ*P* = 100/*n_p* · Σ((*P*_{calcd} − *P*_{exptl})/*P*_{exptl}), where *n_p* is the number of data points and Δ*y* = 1/*n_p* · Σ(*y*_{calcd} − *y*_{exptl}).

**Figure 4.** VLE for toluene (1) + 3-cresol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.**Figure 5.** VLE for toluene (1) + 4-cresol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

and 2-cresol. This is readily understood from the effect of the nonpolar methyl group for the difference of phenol from cresol. In cresols the methyl group causes inference on the formation of hydrogen bonds depending on the position. In the para position the influence is smallest and in the ortho position the strongest. For the system water + 2-cresol the slope of the solubility curve is for both the data of Sidgwick et al.¹⁴ and our data taken from ref 14 and our data different from those of

the systems water + 3-cresol or + 4-cresol. Our experimental LLE data are in good agreement with data from Sidgwick et al. for the system water + 3-cresol and water + 4-cresol.

Calculation

The excess molar volumes *V*^E were calculated using pure substance densities (ρ_{*i*}), measured densities of the mixtures (ρ), mole fraction (*x_i*), and molar mass (*M_i*):

$$V^E/\text{cm}^3 \cdot \text{mol}^{-1} = \frac{x_1 M_1 + x_2 M_2}{\rho} - \frac{x_1 M_1}{\rho_1} - \frac{x_2 M_2}{\rho_2} \quad (3)$$

The excess molar volumes are presented in Figure 16 (values are given in the Supporting Information). The excess molar volumes of the toluene systems are negative over the entire concentration range while the excess molar volume curves of the octane-containing systems are sigmoid with positive values at low cresol concentrations similar to alkane + alcohol systems.¹⁵ The behavior of the excess molar volume curves was described by several authors^{15,16} and is characterized by different effects: positive contributions due to the breaking of hydrogen bonds especially at low phenol or cresol contents (also called chemical interaction); negative contributions due to physical interaction between unlike molecules. In the octane + cresol systems these opposing effects lead to a sigmoid-shaped curve since breaking of hydrogen bonds has an asymmetrical dependence and has larger contribution at low phenol/cresol concentrations. Octane + 2-cresol has more positive excess molar volumes than octane + 3-cresol since the positions of the hydroxyl and methyl group cause stronger steric hindrance in 2-cresol. In addition to the two stated effects, in the toluene systems a negative contribution arises from formation of hydrogen bonds between the π-electrons of toluene and the hydroxyl group. Therefore, the excess molar volume is negative over the entire concentration range. The asymmetrical dependence of the chemical contribution results in shifting the minima to slightly higher phenol/cresol contents. The steric hindrance increases in the order phenol, 4-cresol, 3-cresol, and 2-cresol. Hence, the system toluene + phenol has the largest negative excess molar volume and the system toluene + 2-cresol the smallest.

The Redlich–Kister polynomial equation was used to fit the *V*^E values. The coefficients (*A_i*) were obtained by least-squares regression and are listed together with the standard deviations in Table 3.

$$V_{\text{calcd}}^E/\text{cm}^3 \cdot \text{mol}^{-1} = x_1 x_2 \sum_{i=1}^k A_i (x_1 - x_2)^{i-1} \quad (4)$$

Table 3. Redlich Kister Coefficients (eq 3) and Standard Deviations

system	A_1 (cm ³ ·mol ⁻¹)	A_2 (cm ³ ·mol ⁻¹)	A_3 (cm ³ ·mol ⁻¹)	A_4 (cm ³ ·mol ⁻¹)	s^a (cm ³ ·mol ⁻¹)
toluene + phenol	-1.2413	0.3561	0.0792		0.002
toluene + 2-cresol	-0.6389	0.2660	0.1637		0.003
toluene + 3-cresol	-0.8281	0.2002	0.0855		0.005
toluene + 4-cresol	-0.9125	0.1173	0.0479		0.003
octane + 2-cresol	-0.7699	0.7911	0.3095	1.4761	0.010
octane + 3-cresol	-1.2386	1.4520	-0.3349	0.9012	0.005

^a $s = (1/(n_p - 1) \cdot \sum (V_{\text{calcd}}^E - V_{\text{exptl}}^E)^2)^{1/2}$, where n_p is the number of data points.

Table 4. Pure Component Parameters for the ESD EOS

	ref	c	ϵ/k (K)	b (cm ³ ·mol ⁻¹)	$\epsilon_{\text{HB}}/RT_c$	K_{AB}/ν^*	T_r	ΔP^a (%)	$\Delta \nu^a$ (%)
phenol	this work	1.6503	415.407	34.391	3.0470	0.02936	0.45–0.68	0.13	0.88
2-cresol	this work	1.1983	607.630	42.618	4.3269	0.00098	0.45–0.67	0.66	0.97
3-cresol	this work	1.6317	444.525	41.836	3.3574	0.01201	0.42–0.67	0.57	0.55
4-cresol	this work	2.0758	359.595	40.150	2.2174	0.1517	0.44–0.68	0.54	1.04
octane	20	2.4842	285.211	54.157					
toluene	20	1.9707	332.752	36.227					
water	20	1.0053	427.254	9.411	4.0000	0.100			

^a $\Delta P = 100/n_p \cdot \sum ((X_{\text{calcd}} - X_{\text{exptl}})/X_{\text{exptl}})$, where n_p is the number of data points and X is the vapor pressure P or the liquid density ν .

Table 5. Binary Interaction Parameters and Deviations for the NRTL and UNIQUAC Model

binary systems	fitted data	model	C_{12}^T (K)	C_{21}^T (K)	C_{12}^T	C_{21}^T	ΔP^a (%)	Δy^b	Δx^b
toluene + phenol	this work, 23, 24	NRTL ($\alpha = 0.20$)	857.14	-308.41	-4.3775	2.8430	1.83	0.0051	
		UNIQUAC	369.57	-146.32	-1.6642	0.9573	1.84	0.0051	
toluene + 2-cresol	this work	NRTL ($\alpha = 0.20$)	942.67	-416.16	-5.9518	2.9683	0.78	0.0054	
		UNIQUAC	372.85	-191.68	-2.7563	1.6261	0.75	0.0054	
toluene + 3-cresol	this work, 25	NRTL ($\alpha = 0.20$)	909.79	-303.03	-5.1134	2.2854	2.01	0.0015	
		UNIQUAC	352.72	-149.49	-2.4085	1.3615	1.73	0.0015	
toluene + 4-cresol	this work, 26	NRTL ($\alpha = 0.20$)	443.83	-50.11	1.6730	-1.6607	1.46	0.0042	
		UNIQUAC	108.66	4.74	0.9399	-0.8258	1.48	0.0042	
octane + 2-cresol	this work	NRTL ($\alpha = 0.51$)	707.78	579.79	-2.8088	-1.4006	1.06	0.0019	
		UNIQUAC	286.41	-57.92	0.0066	-0.2871	1.35	0.0014	
octane + 3-cresol	this work	NRTL ($\alpha = 0.40$)	606.70	-244.16	-1.2432	7.3141	2.31	0.0031	
		UNIQUAC	519.83	-266.85	-2.7093	2.3092	1.86	0.0027	
water + phenol	this work, 14, 27	NRTL ($\alpha = 0.20$)	1821.63	-557.19	-5.0783	1.5755	6.35	0.0185	0.0250
		UNIQUAC	467.34	-216.70	0.6408	-0.9202	5.03	0.0189	0.0116
water + 2-cresol	this work	NRTL ($\alpha = 0.40$)	1470.53	439.33	1.8177	-3.1878	0.67	0.0017	0.0030
		UNIQUAC	569.14	-181.64	-1.4436	0.1756	0.86	0.0012	0.0411
water + 3-cresol	this work, 14	NRTL ($\alpha = 0.40$)	1170.12	294.50	4.0054	-2.0853	2.00	0.0036	0.0059
		UNIQUAC	41.34	74.19	-1.5048	1.7006	2.72	0.0033	0.0086
water + 4-cresol	this work, 14	NRTL ($\alpha = 0.20$)	1502.87	-307.39	4.4922	-1.8663	0.61	0.0110	0.0119
		UNIQUAC	243.42	-87.22	0.6966	-0.7445	0.98	0.0105	0.0068
average		NRTL					1.91	0.0056	0.0114
		UNIQUAC					1.86	0.0054	0.0170

^a $\Delta P = 100/n_p \cdot \sum ((P_{\text{calcd}} - P_{\text{exptl}})/P_{\text{exptl}})$, where n_p is the number of data points. ^b $\Delta Z = 1/n_p \cdot \sum ((Z_{\text{calcd}} - Z_{\text{exptl}}))$ where Z represents y or x .

Table 6. Binary Interaction Parameters and Deviations for the ESD EOS

binary systems	fitted data	k_{ij}^C	k_{ij}^T (K ⁻¹)	ΔP^a (%)	Δy^b	Δx^b
toluene + phenol	this work, 23, 24	-0.01053	0.0000106	2.51	0.0070	
toluene + 2-cresol	this work	-0.03536	0.0003213	3.18	0.0084	
toluene + 3-cresol	this work, 25	-0.01417	0.0001157	2.19	0.0042	
toluene + 4-cresol	this work, 26	-0.00838	0.0000210	2.20	0.0062	
octane + 2-cresol	this work	-0.04133	0.0005601	2.46	0.0066	
octane + 3-cresol	this work	-0.03144	0.0004352	2.67	0.0072	
water + phenol	this work, 14, 27	0.04429	0.0001469	5.25	0.0173	0.0280
water + 2-cresol	this work	0.03602	0.0004542	3.13	0.0009	0.0056
water + 3-cresol	this work, 14	0.05225	0.0002751	1.81	0.0035	0.0101
water + 4-cresol	this work, 14	0.04577	0.0001782	5.05	0.0073	0.0095
average				3.05	0.0069	0.0133

^a $\Delta P = 100/n_p \cdot \sum ((P_{\text{calcd}} - P_{\text{exptl}})/P_{\text{exptl}})$, where n_p is the number of data points. ^b $\Delta Z = 1/n_p \cdot \sum ((Z_{\text{calcd}} - Z_{\text{exptl}}))$ where Z represents y or x .

The resulting experimental uncertainty for V^E is $\Delta V^E/\text{cm}^3 \cdot \text{mol}^{-1} = 0.015$ for the measured systems. The standard deviations of experimental and calculated excess molar volumes are always less than this value.

The experimental VLE and LLE data of the binary systems were correlated using binary interaction parameters that are linear temperature-dependent. The excess Gibbs energy models UNIQUAC and NRTL and the ESD EOS were used to describe

the phase behavior. The interaction parameters of the activity coefficient models were determined as follows,

$$C_{ij}/K = C_{ij}^C + C_{ij}^T(T - 273.15 \text{ K}) \quad (5)$$

with $C_{ij} = (u_{ij} - u_{jj})/R$ for UNIQUAC and $C_{ij} = (g_{ij} - g_{jj})/R$ for NRTL. The objective function of Renon et al.¹⁷ was used,

$$Q = \prod_i \sum_i \left(\frac{100}{P_{\text{exptl}}}_i \right)^2 (P_{\text{calcd}} - P_{\text{exptl}})_i^2 + \prod_i \sum_i (100)_i^2 (y_{1,\text{calcd}} - y_{1,\text{exptl}})_i^2 + \prod_i \sum_i (100)_i^2 (x'_{1,\text{calcd}} - x'_{1,\text{exptl}})_i^2 + \prod_i \sum_i (100)_i^2 (x''_{1,\text{calcd}} - x''_{1,\text{exptl}})_i^2 \quad (6)$$

where Π_i denotes a weighting factor, P pressure, and y vapor phase and x' and x'' denote liquid-phase composition.

The pure component parameters of the ESD EOS are listed in Table 4. For the substances water, octane, and toluene the parameter sets of Elliott et al. from ref 20 were used. For phenol, 2-cresol, 3-cresol, and 4-cresol the parameters were determined from vapor pressure and volume of saturated liquid data in the temperature range from melting point to 473.15 K. The vapor pressures were calculated from the Frost Kalkwarf equation.²¹ The volumes of saturated liquid V^L were determined using the Rackett equation modified by Spencer and Danner:²²

$$V^L = \frac{RT_c}{P_c} Z_{RA}^{[1+(1-T_r)^{27}]} \quad (7)$$

R denotes the universal gas constant, T_c and P_c are critical pressure and temperature, and Z_{RA} ²² is a unique constant for each component.

In the association term of the ESD EOS the characteristic association parameter α must be evaluated:

$$\alpha = \frac{\eta}{1 - 1.9\eta} \frac{K_{AB}}{v^*} \left[\exp\left(\frac{\epsilon^{\text{HB}}}{RT}\right) - 1 \right] \quad (8)$$

α is calculated from the pure component parameters energy of hydrogen bonds $\epsilon^{\text{HB}}/RT_c$, volume of hydrogen bonds K_{AB}/v^* , and reduced density $\eta = \rho \cdot b$ with volume parameter b and molar density ρ . The α value is considerably greater for water and decreases in the order phenol, 4- and 3-cresol, and 2-cresol at 333.15 K. This indicates that physically meaningful parameters were obtained.

In cross associating systems (e.g., water + phenol) the characteristic cross association parameter is calculated using the geometric mean solvation rule (also called the Elliott Combining Rule ECR):⁵

$$\alpha_{ij} = \sqrt{\alpha_{ii}\alpha_{jj}} \quad (9)$$

This assumption is only an approximation, but was successfully applied for water + methanol + alkane¹⁸ and amine + alcohol mixtures.¹⁹

The binary interaction parameter k_{ij} of the ESD EOS was also assumed to be linear temperature-dependent.

$$k_{ij} = k_{ij}^C + k_{ij}^T(T - 273.15 \text{ K}) \quad (10)$$

The calculated binary interaction parameters for NRTL, UNIQUAC, and ESD EOS are presented in Tables 5 and 6 together with the deviations for total pressure and vapor composition and for systems with LLE also with the deviation of the liquid compositions.

It is worthy of mention that for the toluene + phenol or + cresols systems negative interaction parameters k_{ij} are obtained, while for the systems water + phenol or + cresols positive interaction parameters k_{ij} are determined. A feasible explanation is that in the toluene systems solvation occurs between the π -electrons of toluene and the hydroxyl group of the cresols. This complexation is not considered in ESD

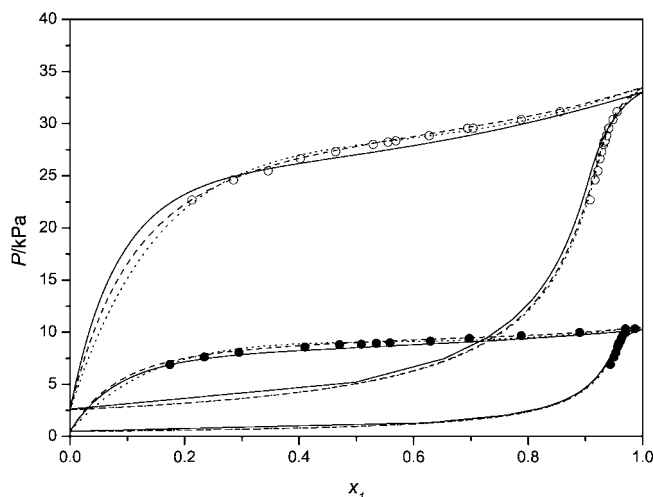


Figure 6. VLE for octane (1) + 2-cresol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

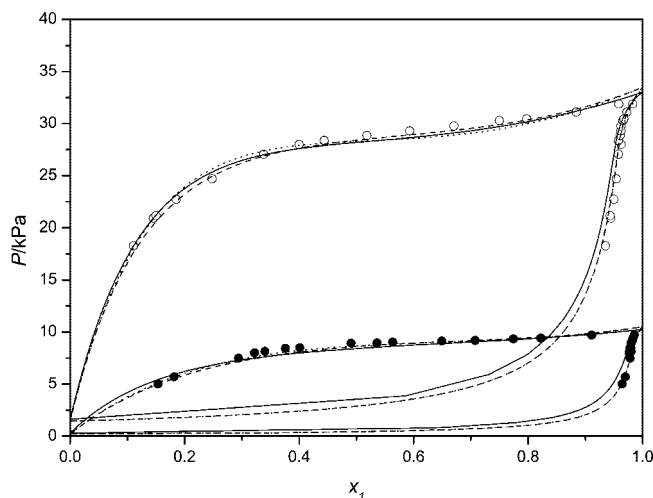


Figure 7. VLE for octane (1) + 3-cresol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

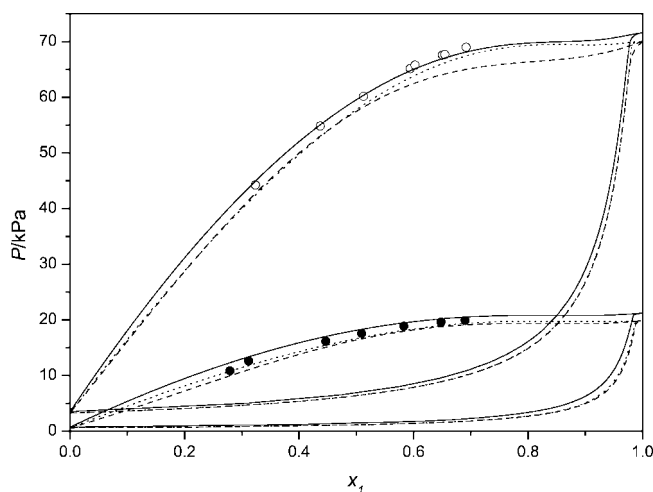


Figure 8. VLE for water (1) + phenol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

EOS since toluene is treated as a nonassociating component and therefore interaction is increased by increasing of dispersive interaction. In the water systems interaction due to solvation may be overestimated by ECR and therefore is compensated by decreasing of dispersive interaction.

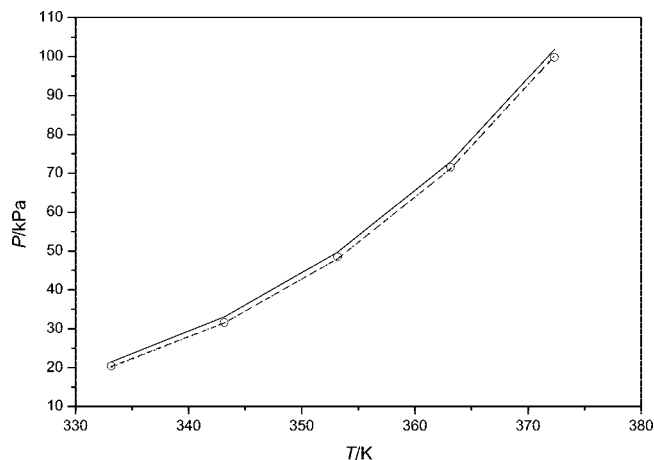


Figure 9. Heteroazeotropic points for water (1) + 2-cresol (2). O, total pressure; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

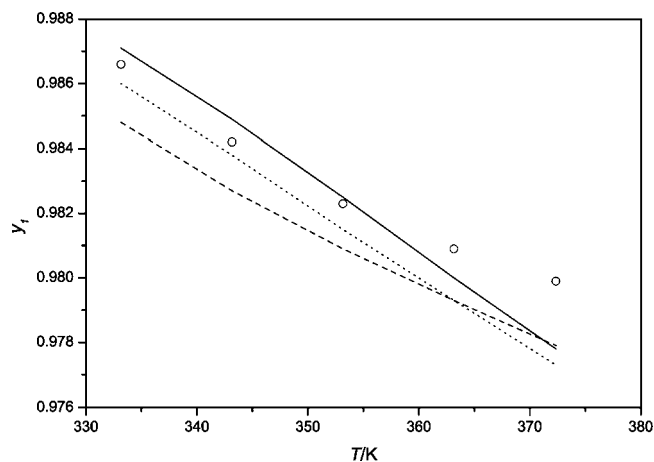


Figure 10. Heteroazeotropic points for water (1) + 2-cresol (2). O, vapor composition; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

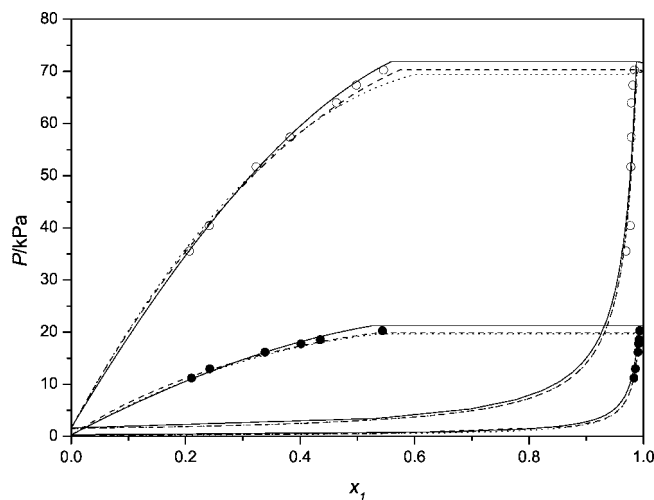


Figure 11. VLE for water (1) + 3-cresol (2). ●, 333.15 K; ○, 363.15 K; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

The results of the activity coefficient models are similar. Therefore, the curves are overlapping for NRTL and UNIQUAC in Figure 2 to Figure 5, Figure 9, and Figure 12. The VLE and VLLE are described by the activity coefficient models with a deviation of ca. 1.9% in total pressure and ca. 0.0055 in vapor composition. The results for the ESD EOS are slightly worse; total pressure is fitted with a deviation of

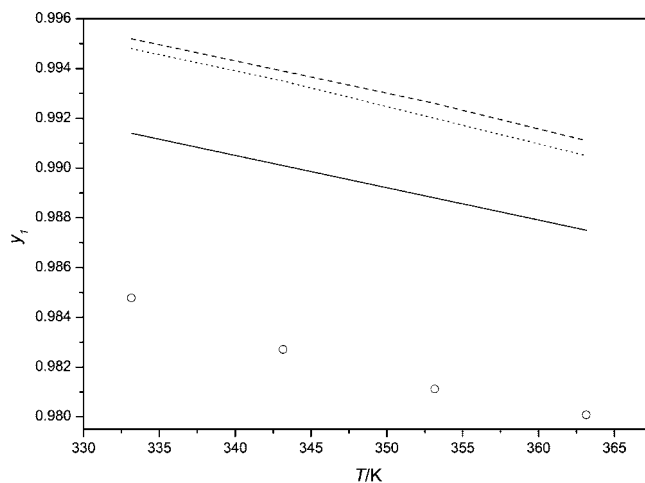


Figure 12. Heteroazeotropic points for water (1) + 4-cresol (2). O, total pressure; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

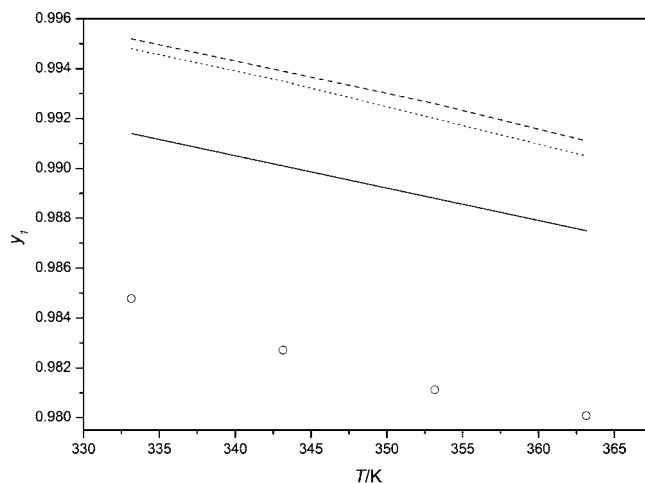


Figure 13. Heteroazeotropic points for water (1) + 4-cresol (2). O, vapor composition; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

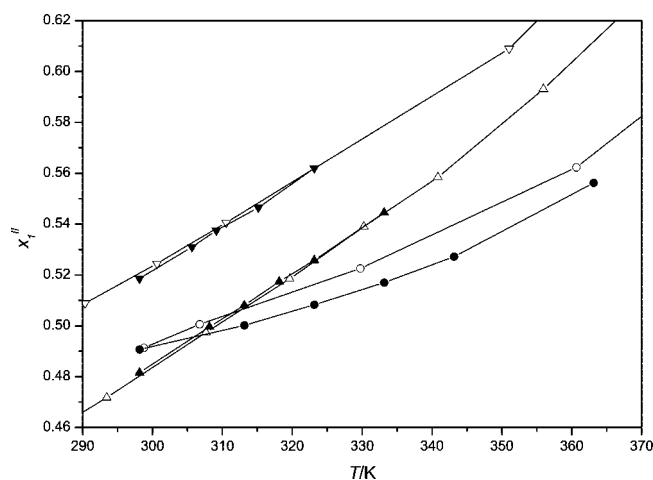


Figure 14. Solubility of water (1) in 2-cresol (2), 3-cresol (2), or 4-cresol (2). Experimental data of this work: ●, 2-cresol; ▲, 3-cresol; ▼, 4-cresol. Sidgwick et al. taken from ref 14: ○, 2-cresol; ▲, 3-cresol; ▽, 4-cresol.

3.05% and vapor composition with 0.0068. The deviations of experimental and calculated total pressures and vapor compositions are summarized and compared in Figures 17 and 18.

The LLE are satisfactorily described by all models. The deviations for both phases are 0.011 for NRTL, 0.017 for

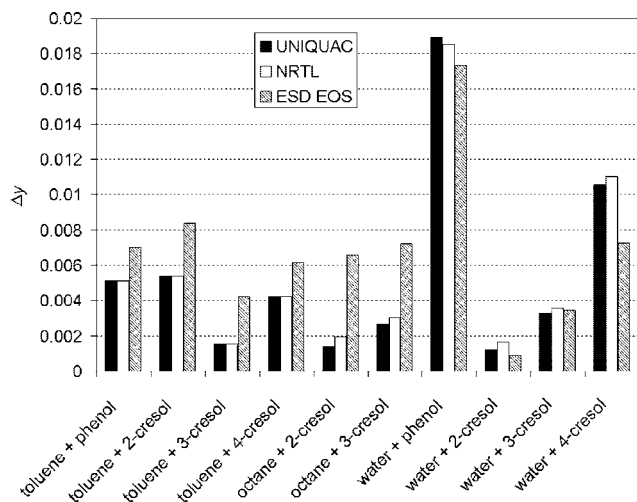


Figure 15. Solubility of water (1) in 2-cresol (2), 3-cresol (2), or 4-cresol (2). Experimental data of this work. ●, 2-cresol; ▲, 3-cresol; ▼, 4-cresol; ---, NRTL; ···, UNIQUAC; —, ESD EOS.

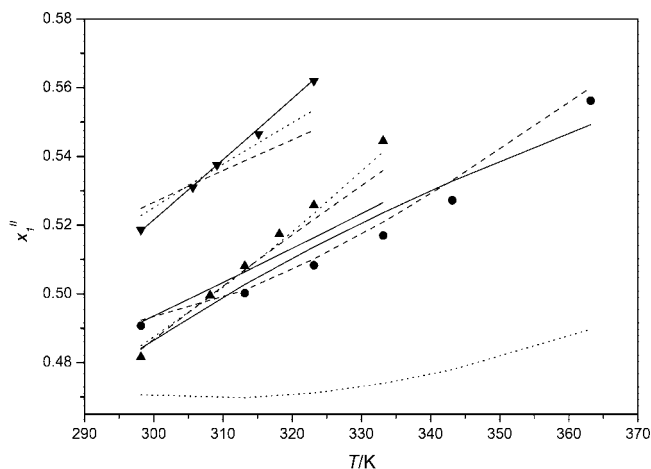


Figure 16. Excess molar volumes. Experimental results ■, toluene (1) + phenol (2) (313.15 K); ●, toluene (1) + 2-cresol (2) (303.15 K); ▲, toluene (1) + 3-cresol (2) (303.15 K); ▼, toluene (1) + 4-cresol (2) (303.15 K); ○, octane (1) + 2-cresol (2) (303.15 K); △, octane (1) + 3-cresol (2) (303.15 K); —, Redlich-Kister polynomial fit (eq 4).

UNIQUAC, and 0.013 for ESD EOS. The deviations of experimental and calculated liquid composition of the cresol-rich phase in the systems water + 2-cresol, + 3-cresol, and + 4-cresol are presented in Figure 15. The description of the system water + 2-cresol with the UNIQUAC model resulted in higher deviations. It has to be mentioned that the NRTL model uses five adjustable parameters (α parameter and two temperature-dependent interaction parameters), the UNIQUAC model uses four adjustable parameters (two temperature-dependent interaction parameters), and ESD EOS uses only two adjustable parameters (one temperature-dependent interaction parameter). Therefore, the activity coefficient models may have more flexibility in fitting the binary data and deliver better results.

Conclusion

The VLE of the systems toluene + phenol, + 2-cresol, + 3-cresol, or + 4-cresol, octane + 2-cresol or + 3-cresol, and water + phenol or + 3-cresol were measured at 333.15 and 363.15 K at reduced pressure using the dynamic method. For the heterogeneous systems water + 2-cresol and water +

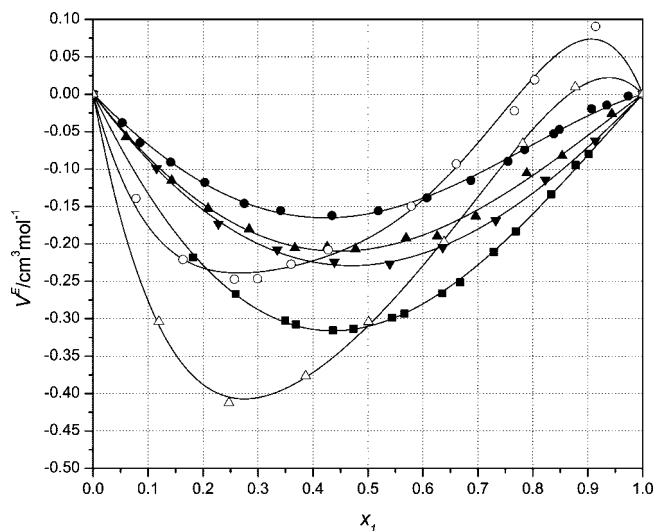


Figure 17. Deviations between experimental and calculated total pressures for NRTL, UNIQUAC, and ESD EOS.

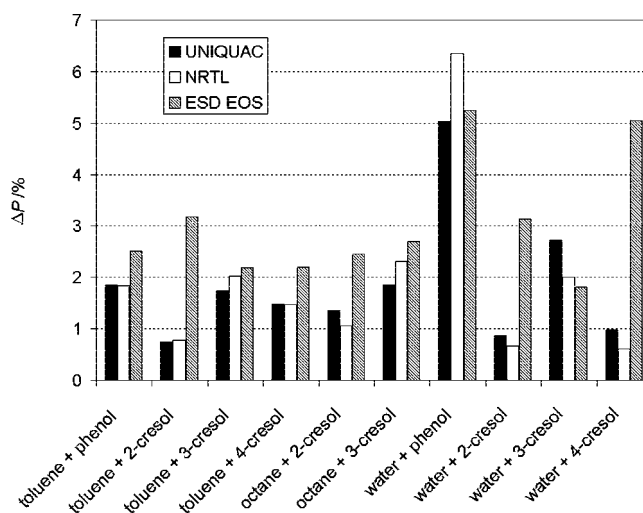


Figure 18. Deviations between experimental and calculated vapor composition for NRTL, UNIQUAC, and ESD EOS.

4-cresol the VLE were determined from 333.15 to 363.15 K in steps of 10 K. The solubility of water in 2-cresol, 3-cresol, or 4-cresol was measured by photometric turbidity titration at ambient pressure. For composition analysis of the VLE samples the densities of mixtures for the homogeneous systems toluene + phenol, + 2-cresol, + 3-cresol, or + 4-cresol and octane + 2-cresol or + 3-cresol were determined and the excess molar volumes were calculated. Binary interaction parameters were determined for the models NRTL, UNIQUAC, and ESD EOS. Available literature data were included in data fitting. The total deviations of all systems for total pressure were 1.91% for NRTL, 1.86% for UNIQUAC, and 3.05% for ESD EOS. The deviations of vapor composition were 0.0056 for NRTL, 0.0054 for UNIQUAC, and 0.0068 ESD EOS (given as mole fraction). The LLE were described with a deviation of 0.011 for NRTL, 0.017 for UNIQUAC, and 0.013 for ESD EOS. The UNIQUAC and the NRTL model yielded satisfactory results. The ESD EOS performed slightly worse than the activity coefficient models for the binary systems, despite a term for association being included in it but with a lower number of adjustable parameters.

Acknowledgment

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Supporting Information Available: Experimental results of VLE determination, solubility measurement, density measurement, and calculated excess molar volumes. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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